MODULE 22 · DOMAIN 8 - TOOLS AND MCPS

Built-in Tool, Custom Tool, Skill or MCP Server?

Four ways to extend Claude, what each one is for, the signals that pick between them, and when to extend what exists instead of building.

DOMAIN 8 of 8 EXAM WEIGHT 10.6% MODULE 22 of 36

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A workshop has tools on the wall, gadgets you made, a manual on the shelf and a shared supplier. **Reach for the one the job needs.**

The four options

OPTION	WHAT IT IS	WHO MAINTAINS IT
Built-in tool	Anthropic-defined: web search, web fetch, code execution, bash, text editor, memory	Anthropic
Custom tool	Your function, defined in your app's request	You, inside one app
Skill	Packaged instructions, scripts and resources, loaded on demand	You, shared as a folder or plugin
MCP server	A standard interface to a capability, reused across hosts	You, run as a service

ON THE EXAM

Anthropic: "MCP connects Claude to data; Skills teach Claude what to do with that data."

Skills, in a minute

- **A folder with SKILL.md:** a name and a description, then instructions, plus optional scripts and files.
- **Progressive disclosure:** metadata always loaded; body when triggered; resources as needed.
- **The description decides when it triggers,** so it is written like a tool description.
- **Where they run:** Claude Code, the Agent SDK, claude.ai, and the API with the code execution tool.
- **Custom Skills do not sync across surfaces;** share them through plugins or the API.

> THE TAKEAWAY
>
> A Skill costs almost nothing until it is needed. That is why many Skills can sit ready at once.

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Signals that pick the option

IF THE QUESTION SAYS	LEAN TOWARDS
Search the web, fetch a page, run code, edit files	A built-in tool
Call our own system, from this one application	A custom tool
Teach a procedure, house style or expertise, used repeatedly	A Skill
Many applications or teams need the same system	An MCP server
Must happen every time, deterministically	Code or a hook, not any of these

THE TIEBREAKER

Name what is missing first: an action, access to a system, know-how, or reuse. Then pick.

Extend or build?

EXTEND WHAT EXISTS

Use the platform

- A built-in tool already does it
- An existing MCP server or connector covers the system
- A published Skill fits the task
- Maintained and improved by someone else

BUILD YOUR OWN

Only when you must

- Your private system or data
- Your rules and permissions
- Nothing existing meets the requirement
- You accept owning it for its life

ON THE EXAM

Rule five: use what the platform gives you. Building is a life-cycle commitment, not a weekend.

Skill or subagent? Skill or hook?

SKILL

Know-how, in context

- Teaches how to do a task
- Runs in the current conversation
- Model decides when it applies
- Guidance, like any prompt

SUBAGENT OR HOOK

Isolation or enforcement

- Subagent: a separate context for a task
- Returns a summary, keeps the main context lean
- Hook: code that runs every time
- Enforces, whatever the model decides

THE TIEBREAKER

Anthropic: use Skills to teach expertise; use subagents for independent task execution.

Copperlet's choices

NEED	CHOICE	WHY
Agency-ops agent reads public guidance pages	Built-in web fetch	Already exists, maintained
Tenant assistant books a repair slot	Custom tool	One app, its own rules
Property data for three hosts	MCP server	Reuse across hosts
How Copperlet writes arrears summaries	Skill	House style, used repeatedly
Block edits to secrets in Claude Code	Hook	Must happen every time

THE TAKEAWAY

One company, five needs, five different answers. The need decides, not habit.

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> A research assistant built on the Messages API must look up recent news articles about companies a user names. The team plans to build and host its own web-scraping service and expose it as a custom tool.
>
> No private data is involved.

What should the team do?

A. Build the scraping service, since custom tools give the most control

B. Use the built-in web search and web fetch tools

C. Write a Skill that explains how to search the web

D. Paste recent news into the system prompt each morning

◆ PAUSE AND DECIDE

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What should the team do?

A Build the scraping service, since custom tools give the most control

Builds and maintains what already exists.

B Use the built-in web search and web fetch tools

Exists, maintained, fits the need.

C Write a Skill that explains how to search the web

Know-how, but no ability to search.

D Paste recent news into the system prompt each morning

Stale, partial and costly.

THE SIGNAL

"Look up recent web content" and "no private data": a built-in tool already does it.

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> A bank's agents in Claude Code, a customer chatbot and a reporting agent all need read access to the same internal accounts system. Each team is about to write its own integration.
>
> Access rules must be consistent.

Which option fits best?

A. A custom tool in each application, following a shared design document

B. A Skill that describes the accounts API in detail

C. A built-in tool, configured with the accounts system's URL

D. One MCP server for the accounts system, used by all three hosts

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Which option fits best?

A A custom tool in each application, following a shared design document

Three copies, three sets of rules.

B A Skill that describes the accounts API in detail

Know-how without enforced access.

C A built-in tool, configured with the accounts system's URL

Built-in tools do not reach private systems.

D One MCP server for the accounts system, used by all three hosts

One integration, one set of rules.

THE SIGNAL

"Same system, several hosts, consistent rules": an MCP server.

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> A consultancy wants Claude to produce client reports in its house format: a set structure, a style guide, a chart script and a template file. Consultants use Claude in Claude Code and in claude.ai, many times a week.
>
> Today they paste the instructions into each conversation.

What is the best option?

A. A custom tool that returns the style guide as text

B. An MCP server that serves the template file

C. A Skill bundling the instructions, script and template, loaded when relevant

D. A longer system prompt in every application

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What is the best option?

A A custom tool that returns the style guide as text

A tool for something that is know-how.

B An MCP server that serves the template file

Serves one file; not the procedure.

C A Skill bundling the instructions, script and template, loaded when relevant

Packaged know-how, on demand.

D A longer system prompt in every application

Always loaded, and not portable.

THE SIGNAL

Repeated procedure plus a script and a template: that is exactly what a Skill packages.

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> A single internal web app lets staff ask Claude about stock levels in the company's warehouse database. No other application needs this, and the database already has a small internal API.
>
> The team wants the simplest sound design.

Which option fits best?

A. A custom tool in the web app that calls the internal API

B. A company-wide MCP server with OAuth and a marketplace listing

C. A Skill describing how to query the warehouse database

D. The built-in code execution tool, connecting to the database directly

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Which option fits best?

A A custom tool in the web app that calls the internal API

One app, one function: simplest sound design.

B A company-wide MCP server with OAuth and a marketplace listing

Reuse infrastructure nobody needs yet.

C A Skill describing how to query the warehouse database

Know-how without access.

D The built-in code execution tool, connecting to the database directly

Sandboxed, and the wrong place for credentials.

THE SIGNAL

"No other application needs this": a custom tool. Promote it to MCP when reuse appears.

> An engineering team wants Claude Code to always run the formatter after editing a file. They wrote a Skill describing the formatting step, but it is applied only some of the time.
>
> The team wants it every time.

What should they use?

A. A longer, more emphatic Skill description

B. An MCP server that offers a format tool

C. A line in CLAUDE.md asking Claude to always format

D. A PostToolUse hook on edits that runs the formatter

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What should they use?

A. A longer, more emphatic Skill description
Still the model's choice.

B. An MCP server that offers a format tool
Still optional to call.

C. A line in CLAUDE.md asking Claude to always format
Context, not enforcement.

D. A PostToolUse hook on edits that runs the formatter
Runs every time, deterministically.

THE SIGNAL

"Every time" means a hook. Skills, tools and CLAUDE.md all leave the choice to the model.

> A team wants Claude Code to investigate failing tests across a large codebase. The investigation reads hundreds of files, and the team's main session keeps running out of useful context afterwards.
>
> Only the conclusion matters to the main session.

What fits best?

A. A Skill that explains how to investigate failing tests

B. A subagent that investigates in its own context and returns a summary

C. An MCP server that exposes the test runner

D. A hook that runs the tests after every edit

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What fits best?

A A Skill that explains how to investigate failing tests

Runs in the same, overloaded context.

B A subagent that investigates in its own context and returns a summary

Isolation; only the conclusion returns.

C An MCP server that exposes the test runner

Access, not isolation.

D A hook that runs the tests after every edit

Runs tests; does not investigate.

THE SIGNAL

"Reads hundreds of files" and "only the conclusion matters": a subagent.

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> A support assistant must create tickets in the company's ticketing system, and agents must follow the company's specific triage procedure when doing so. Two other teams already use an existing ticketing MCP server.
>
> The team is deciding what to add.

Which two choices fit best? Choose two.

- A Write a new custom tool that duplicates the MCP server's functions
- B Use a built-in tool to call the ticketing system's URL
- C Connect the assistant to the existing ticketing MCP server
- D Package the triage procedure as a Skill
- E Put the whole procedure into each tool's description

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Which two choices fit best? Choose two.

A Write a new custom tool that duplicates the MCP server's functions

Duplicates what exists.

B Use a built-in tool to call the ticketing system's URL

No built-in tool for private systems.

C Connect the assistant to the existing ticketing MCP server

Access: reuse what exists.

D Package the triage procedure as a Skill

Know-how: a Skill.

E Put the whole procedure into each tool's description

Bloats every tool call.

THE SIGNAL

Access plus know-how: MCP for the data, a Skill for what to do with it.

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> A team built a Skill in claude.ai for drafting board papers. Engineers expect to find it in their Claude Code sessions too, but it is not there.
>
> Nothing is broken in claude.ai.

What explains this, and what should they do?

A Skills only work in claude.ai, so the engineers cannot use it

B Custom Skills do not sync across surfaces; share it in a plugin

C The Skill's description is too short, so Claude Code ignores it

D Claude Code needs the Skill rewritten as an MCP server

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What explains this, and what should they do?

A Skills only work in claude.ai, so the engineers cannot use it

Claude Code supports Skills.

B Custom Skills do not sync across surfaces; share it in a plugin

The documented behaviour, and its fix.

C The Skill's description is too short, so Claude Code ignores it

It was never installed there.

D Claude Code needs the Skill rewritten as an MCP server

Wrong type of extension.

CURRENCY CHECK

Anthropic: "Custom Skills do not sync across surfaces." Distribute them deliberately.

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Module 22 in five lines

- **Built-in tool** when Anthropic already provides the action.
- **Custom tool** for your system, in one application.
- **Skill** for repeated know-how; **MCP server** for a system many hosts reuse.
- **Every time?** A hook or code. **Context-heavy task?** A subagent.
- **Extend before you build;** building means owning it for life.

COMING UP

Next, Part 6: workflows or agents?